

TQC2

Probability and Statistics II

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A booster club printed 500 raffle tickets at \$5 each. Each ticket has a 0.02 chance of winning \$100 and a 0.05 chance of winning \$20; otherwise it wins nothing. The drawing is held at halftime. What is the expected net gain for a person who buys one ticket?

- A. +\$2
- B. -\$2
- C. \$60
- D. \$3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

Wait times at a clinic have mean $\mu = 30$ minutes and standard deviation $\sigma = 12$ minutes. The distribution of individual wait times is right-skewed. An auditor takes a random sample of 36 visits. What is the standard error of the sample mean?

- A. 2 minutes
- B. It cannot be found because the population is skewed.
- C. 12 minutes
- D. About 0.33 minutes

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A population has $\sigma = 12$. What sample size gives a standard error of 1.5 for the sample mean?

- A. 8
- B. 144
- C. 64
- D. 18

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A snack plant that runs two shifts fills bags with a mean of 250 g and a standard deviation of 4 g. An inspector weighs one bag at 259 g. The plant's packaging supplier changed in May. Is this bag unusual?

- A. $z = 2.25$, so it is unusual.
- B. $z \approx 0.44$, so it is typical.
- C. $z = 9$, so it is unusual.
- D. $z = 2.25$, but it is not unusual until z passes 3.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

5.

Two technicians collected data over 3 weeks and built a 95% confidence interval of (48.2, 53.8) grams for the mean weight of a part. Which interpretation is correct?

- A. The population mean is definitely between 48.2 and 53.8 grams.
- B. 95% of individual parts weigh between 48.2 and 53.8 grams.
- C. There is a 95% chance the sample mean is in the interval.
- D. We are 95% confident the population mean weight is between 48.2 and 53.8 grams.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

6. [TRANSFER — reversed stem / different variable]

Which change will make a confidence interval for a mean narrower, all else equal?

- A. Using a smaller sample to reduce noise
- B. Raising the confidence level from 95% to 99%
- C. Nothing; the width is fixed by σ .
- D. Increasing the sample size

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

7.

A product team spent \$4,000 on a pilot to test whether a new checkout page raises conversion. With $\alpha = 0.05$, the test gives $p = 0.03$. The page was designed by an outside agency. What should the team conclude?

- A. Reject H_0 ; there is evidence the new page raises conversion.
- B. Fail to reject H_0 , because the p-value is small.
- C. The new page is proven to raise conversion.
- D. There is a 3% chance H_0 is true.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

8. [TRANSFER — reversed stem / different variable]

A test uses $\alpha = 0.01$ and gives $p = 0.03$. What is the decision?

- A.** Reject H_0 , because 0.03 is less than 0.05.
- B.** Reject H_0 , because 0.03 is less than $1 - 0.01 = 0.99$.
- C.** Fail to reject H_0 , because 0.03 is greater than 0.01.
- D.** Accept H_0 ; it is now proven true.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A chemical lab tests H_0 : a batch meets the purity standard. Each batch costs \$12,000 to make. The lab rejects H_0 and discards a batch that actually met the standard. The lab manager has run the process for a decade. What happened?

- A.** No error, because rejecting H_0 is always the safe choice.
- B.** A Type I error
- C.** A Type II error
- D.** Low power ($1 - \beta$)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A pharmacist tests H_0 : the mean dose is 500 mg against H_a : the mean dose is not 500 mg. What would a Type II error mean?

- A.** Concluding the dose is wrong when it actually is 500 mg.
- B.** Rejecting H_0 whenever $p < \alpha$.
- C.** Concluding the dose is fine when it actually is not 500 mg.
- D.** Weighing a tablet on a broken scale.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A training vendor that charges per seat measures the typing speed of the same 20 employees before and after a two-day course. Which test fits the data?

- A.** Independent two-sample t-test
- B.** Paired t-test
- C.** One-way ANOVA
- D.** Chi-squared test of independence

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A pollster surveyed 200 voters during a rainy week: 120 favor a transit plan and 80 oppose it. Of the 200, 90 live in urban areas and 110 in rural areas. Under independence, what is the expected count of urban voters who favor the plan?

- A. 45
- B. 60
- C. 36
- D. 54

Target word _____	The one fact that decides _____	Answer _____	☐ sure ☐ unsure ☐ guess
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STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A booster club ~~printed 500 raffle tickets~~ at \$5 each. Each ticket has a 0.02 chance of winning \$100 and a 0.05 chance of winning \$20; otherwise it wins nothing. ~~The drawing is held at halftime.~~ What is the expected net gain for a person who buys one ticket?

Target word: expected net gain **Decides:** Expected prize = $0.02(100) + 0.05(20) = \$3$; subtract the \$5 cost.

A	Inverted operation	Subtracts in the wrong direction.
B	Correct	$\$3 - \$5 = -\$2$ per ticket.
C	Adjacent category	Unweighted average of the two prizes, ignoring the probabilities.
D	Dropped element	Expected prize, but the \$5 cost is dropped.

2. Answer A Concept 2

Wait times at a clinic have mean $\mu = 30$ minutes and standard deviation $\sigma = 12$ minutes. ~~The distribution of individual wait times is right-skewed.~~ An auditor takes a random sample of 36 visits. What is the standard error of the sample mean?

Target word: standard error **Decides:** $SE = \sigma / \sqrt{n} = 12 / 6$.

A	Correct	$12 / \sqrt{36} = 2$.
B	Camouflage	SE does not depend on shape, and with $n = 36$ the CLT still applies.
C	Wrong stem number	That is σ for individual waits.
D	Dropped element	Divides by $n = 36$ and drops the square root.

3. Answer C Concept 2 · transfer

A population has $\sigma = 12$. What sample size gives a standard error of 1.5 for the sample mean?

Target word: solve for n **Decides:** $1.5 = 12 / \sqrt{n}$, so $\sqrt{n} = 8$.

A	Dropped element	That is $\sqrt{n}$; it still has to be squared.
B	Confusion set	That is σ^2 , the variance.
C	Correct	$\sqrt{n} = 8$, so $n = 64$.
D	Inverted operation	Multiplies 12×1.5 instead of dividing.

4. Answer A Concept 3

A snack plant ~~that runs two shifts~~ fills bags with a mean of 250 g and a standard deviation of 4 g. An inspector weighs one bag at 259 g. ~~The plant's packaging supplier changed in May.~~ Is this bag unusual?

Target word: z-score and judgment **Decides:** $z = (259 - 250) / 4 = 2.25$.

A	Correct	More than 2 standard deviations above the mean.
B	Inverted operation	Divides σ by the difference: $4 / 9$.
C	Dropped element	Drops the division by σ .
D	Adjacent category	Borrows a stricter outlier cutoff; the usual line is 2.

5. Answer D Concept 4

~~Two technicians collected data over 3 weeks~~ and built a 95% confidence interval of (48.2, 53.8) grams for the mean weight of a part. Which interpretation is correct?

Target word: interpret a CI **Decides:** A CI is about the population mean, and the 95% describes the method.

A	Absolute language	About 5% of such intervals miss μ .
B	Confusion set	A CI for a mean says nothing about the spread of individual parts.
C	Adjacent category	The sample mean is the interval's center, so it is always inside.
D	Correct	The interval gives plausible values for μ from a method that works in about 95% of samples.

6. Answer D Concept 4 • transfer

Which change will make a confidence interval for a mean narrower, all else equal?

Target word: what narrows a CI **Decides:** Width is driven by $z^* \times \sigma / \sqrt{n}$.

A	Confusion set	A smaller n increases the standard error.
B	Inverted operation	Higher confidence uses a larger z^* , which widens the interval.
C	Absolute language	Both n and confidence level change the width.
D	Correct	A larger n shrinks $\sigma / \sqrt{n}$.

7. Answer A Concept 5

A product team ~~spent \$4,000 on a pilot~~ to test whether a new checkout page raises conversion. With $\alpha = 0.05$, the test gives $p = 0.03$. ~~The page was designed by an outside agency.~~ What should the team conclude?

Target word: decision rule **Decides:** $p = 0.03 \leq \alpha = 0.05$.

A	Correct	p is below α .
B	Inverted operation	A small p -value is evidence against H_0 .
C	Absolute language	Tests give evidence, not proof.
D	Confusion set	p is the chance of data this extreme if H_0 were true, not the chance H_0 is true.

8. Answer C Concept 5 • transfer

A test uses $\alpha = 0.01$ and gives $p = 0.03$. What is the decision?

Target word: decision with a stricter α **Decides:** $0.03 > 0.01$.

A	Adjacent category	Uses the habitual 0.05 instead of the stated α .
B	Inverted operation	Compares p with the confidence level instead of α .
C	Correct	p must be at or below α to reject.
D	Absolute language	Failing to reject does not prove H_0 .

9. Answer B Concept 6

A chemical lab tests H_0 : a batch meets the purity standard. ~~Each batch costs \$12,000 to make.~~ The lab rejects H_0 and discards a batch that actually met the standard. ~~The lab manager has run the process for a decade.~~ What happened?

Target word: name the error **Decides:** H_0 was true and was rejected.

A	Absolute language	Rejecting a true H_0 is an error whatever the cost.
B	Correct	Rejecting a true null hypothesis is a Type I error.
C	Inverted operation	Type II is failing to reject a false H_0 .
D	Confusion set	Power concerns detecting a false H_0 , not rejecting a true one.

10. Answer C Concept 6 · transfer

A pharmacist tests H_0 : the mean dose is 500 mg against H_a : the mean dose is not 500 mg. What would a Type II error mean?

Target word: describe a Type II error in context **Decides:** Type II = fail to reject a false H_0 .

A	Inverted operation	That is a Type I error.
B	Adjacent category	That is the decision rule, not an error.
C	Correct	H_0 is false but not rejected.
D	Confusion set	Measurement error is not a hypothesis-testing error type.

11. Answer B Concept 7

A training vendor ~~that charges per seat~~ measures the typing speed of the same 20 employees before and after a ~~two day course~~. Which test fits the data?

Target word: choose the test **Decides:** Each employee is measured twice.

A	Dropped element	Drops the pairing and treats the scores as two separate groups.
B	Correct	Before-and-after on the same people gives paired differences.
C	Adjacent category	ANOVA compares three or more group means.
D	Confusion set	Chi-squared is for categorical counts, not speeds.

12. Answer D Concept 8

A pollster surveyed 200 voters ~~during a rainy week~~: 120 favor a transit plan and 80 oppose it. Of the 200, 90 live in urban areas and 110 in rural areas. Under independence, what is the expected count of urban voters who favor the plan?

Target word: expected cell count **Decides:** (row total $\times$ column total) / grand total = $90 \times 120 / 200$.

A	Wrong stem number	Half of the 90 urban voters; ignores the 60% support rate.
B	Adjacent category	Splits the 120 supporters evenly across areas.
C	Confusion set	That is the urban-oppose cell: $90 \times 80 / 200$.
D	Correct	$10,800 / 200 = 54$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Expected value of a discrete random variable	$E(X) = \sum x \cdot P(x)$, weighting every outcome, including costs, by its probability.	Averaging outcomes without weights, or forgetting the cost to play.	A table of payoffs with probabilities.
2	Central Limit Theorem and standard error	The sample mean has standard error $\sigma / \sqrt{n}$ and is approximately normal for large n , even when the population is skewed.	Dividing by n instead of $\sqrt{n}$, or thinking skew blocks the CLT for a large sample.	A population σ , a sample size, and “the sample mean.”
3	Identifying unusual values	$z = (x - \mu) / \sigma$; a value more than about 2 standard deviations from the mean is usually called unusual.	Forgetting to divide by σ , or waiting for $z > 3$.	A single observation compared with a known mean and SD.
4	Confidence intervals	A 95% CI gives a range of plausible values for the parameter; the method captures the parameter in about 95% of samples. Larger n narrows it; higher confidence widens it.	Saying 95% of individual values fall in the interval, or that more confidence narrows it.	“Interpret the interval” or “which change makes it narrower?”
5	P-value decisions	Reject H_0 when $p \leq \alpha$; otherwise fail to reject. Neither decision proves a hypothesis.	Using 0.05 by habit when the stem sets another α , or saying H_0 is “accepted.”	A p-value and a significance level.
6	Type I and Type II errors	Type I = rejecting a true H_0 ; Type II = failing to reject a false H_0 .	Swapping the two, or naming an error by its cost rather than its logic.	A description of a wrong conclusion in context.
7	Choosing the right test	Same subjects measured twice → paired t ; two separate groups → independent t ; three or more means → ANOVA; categorical counts → chi-squared.	Treating paired data as two independent samples.	“Before and after” on the same people.
8	Expected counts in a chi-squared test	Expected count = (row total $\times$ column total) / grand total.	Splitting the total evenly, or forgetting to divide by the grand total.	A two-way table and “expected count.”

Course scope (public description): Probability and Statistics II covers the knowledge and skills necessary to apply random variables, sampling distributions, estimation, and hypothesis testing, and to use appropriate technology to model and solve real-life problems. It provides tools for the science of analyzing and interpreting data and includes statistical variability and its sources and the role of randomness in statistical inference. Topics include discrete and continuous random variables; expected values; the Central Limit Theorem; the identification of unusual samples; population parameters; point estimates; confidence intervals; influences on accuracy and precision; hypothesis testing; and statistical tests (z mean, z proportion, one sample t , paired t , independent t , ANOVA, chi-squared, and significance of correlation). Calculus II and Probability and Statistics I are prerequisites for this course.